Closed-loop optogenetic control of the dynamics of neural activity in non-human primates

Paper: Zaaimi et al. (2023 (online 2022)). Mouse slices, anaesthetized primates and modelling.

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Page references below use PDF page numbers, including front matter.

To which general question does this paper relate?

Direct evidence for feedback control of neural oscillations and seizure-like activity. Also supplies a concrete rate-model structure compatible with the notebook.

How does it answer the general questions?

Measured and controlled. LFP-dependent, phase-shifted optical stimulation enhanced or attenuated oscillations and modulated chemically induced seizure-like bursts. The same excitatory actuator could have opposite network effects depending on feedback phase (PDF pp. 2–11).

Objective. The experiments target oscillatory dynamics and seizure-burst duration, not Parkinsonian beta or demonstrated clinical recovery. Phase and measurement filtering matter; reducing stimulation intensity alone does not describe the intervention.

Model mapping. Their two-population model is

$$\dot{E} = \frac{-E + \sigma(aE - bI + S + P)}{\tau_e} + \eta_e, \quad \dot{I} = \frac{-I + \sigma(cE - dI + Q)}{\tau_i} + \eta_i.$$

After equilibrium centring, take $x = (E, I)$, linear decay $(-1/\tau_e, -1/\tau_i)$, signed input rows $(a, -b)$ and $(c, -d)$, and $u = S$ inside the excitatory sigmoid. Intrinsic delays are zero. The logistic has slope at most $1/4$, so scaled nonlinearities have bounds $1/(4\tau_i)$. LFP filtering and the feedback FIR filter require additional dynamics; numerical parameters are in the separately referenced Supplementary Table 2 (PDF pp. 14–15).

What papers does it cite that could provide more information?

- Nicholson et al. (2018), *Analogue closed-loop optogenetic modulation of hippocampal pyramidal cells dissociates gamma frequency and amplitude* (ref. 8, PDF p. 15).
- Wang et al. (2014), *Dynamic mechanisms of neocortical focal seizure onset*: model background (ref. 15, PDF p. 15).